Microeconomics Analysis: Linear Algebra

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Course: Introduction

Two lectures per week (one lecture on week 8):

- Tuesday, 10:30-12:00, Room 1, ES
- Wednesday 11:00-12:30, Seminar Room C, MRB

Classes trail lectures by one week.

Classes are delivered by lecturers on their material. Your 'tutor' is the person who will write your TMS report.

Classes schedule: Thursday 14:00, 15:45 and Friday 15:45, Seminar Room A.

Problem sets for each week are going to appear on Canvas. You are expected to solve one problem indicated in the assignment and submit it for marking via Canvas.

The deadline for submission is 8am on Wednesday on the week of the class.

The rest of the problem set is going to be discussed in the class.

Course: Structure

- Week 1: Linear Algebra
- Week 2: Multivariate Calculus
- Weeks 3-4: Decision Theory
- Week 5: Static Optimisation
- Week 6: Information Economics
- Weeks 7-8: Dynamic Optimisation and Search

Linear Algebra: Introduction

Lectures: Week 1

Some references:

- Math Workbook (University of Oxford)
- Mathematics for Economists, Simon and Blume
- Mathematics for Economists, Pemberton and Rau

Linear Algebra

Outline

1. Basic Concepts: Vector Space, Linear Independence and Basis
2. Matrices
3. Basic Operations
4. Determinant, Trace and Rank – Relation to Span
5. Eigenvalues and Eigenvectors

Applications:

A. Linear Systems

B. Linear Transformations

Space of n-dimentional Vectors

We are going to look at a space of all n-dimensional vectors, $\mathbb{R}^n$, either column or row, they will be indicated in **bold**.

Points in $\mathbb{R}$ are referred as scalars.

We define two operations: addition and multiplication by a scalar. For all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$

$$\mathbf{x} + \mathbf{y} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{pmatrix}$$

and for all $\mathbf{x} \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$

$$\lambda \mathbf{x} = \begin{pmatrix} \lambda x_1 \\ \vdots \\ \lambda x_n \end{pmatrix}$$

Axioms

There are some natural properties:

- $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$
- $\mathbf{x} + (\mathbf{y} + \mathbf{z}) = (\mathbf{x} + \mathbf{y}) + \mathbf{z}$
- $\mathbf{x} + \mathbf{0} = \mathbf{x}$
- $\mathbf{x} + (-\mathbf{x}) = \mathbf{0}$
- $1\mathbf{x} = \mathbf{x}$
- $\lambda(\mathbf{x} + \mathbf{y}) = \lambda\mathbf{x} + \lambda\mathbf{y}$
- $(\lambda + \mu)\mathbf{x} = \lambda\mathbf{x} + \mu\mathbf{x}$
- $(\lambda\mu)\mathbf{x} = \lambda(\mu\mathbf{x}) = \mu(\lambda\mathbf{x})$

Examples in this course are vectors of: demands, endowments, prices, security investments, etc.

Dot Product

Another useful concept is **dot product** or **scalar product** of two vectors.

For all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$

$$\mathbf{x} \cdot \mathbf{y} = x_1 y_1 + x_2 y_2 + \ldots x_n y_n$$

Example: if $\mathbf{x}$ is a demand vector and $\mathbf{p}$ is a vector of prices, then $\mathbf{p} \cdot \mathbf{x}$ is consumer expenditure.

A **norm** of a vector $\mathbf{x}$ is $||\mathbf{x}|| = \sqrt{\mathbf{x} \cdot \mathbf{x}}$.

Linear Independence, Definition

Consider a space of n-dimensional vectors, $\mathbb{R}^n$.

- A vector $\mathbf{x}$ is a **linear combination** of a collection of vectors $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_m$ if there exist real numbers $\alpha_1, \alpha_2, \ldots, \alpha_m$ such that

$$\mathbf{x} = \alpha_1 \mathbf{x}_1 + \alpha_2 \mathbf{x}_2 + \ldots + \alpha_m \mathbf{x}_m$$

- A collection of vectors $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_m$ is **linearly independent** if none of them is a linear combination of the others.

Linear Independence, Examples

Vectors

$$\mathbf{x}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \ \mathbf{x}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

are linearly independent, as $\mathbf{x}_1 \neq \alpha \mathbf{x}_2$.

Vectors

$$\mathbf{y}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \ \mathbf{y}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \ \mathbf{y}_3 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

are not linearly independent, as $\mathbf{y}_3 = 2\mathbf{y}_1 + \mathbf{y}_2$.

Intuition: you cannot make a menu consisting of French Fries out of a Big Mac. But you can make a menu consisting one two Big Macs and French Fries out of solo menus.

Basis

A collection of linearly independent vectors $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_m$ is called **basis** of $\mathbb{R}^n$ is any other vector $\mathbf{x} \in \mathbb{R}^n$ is a linear combination of these vectors,

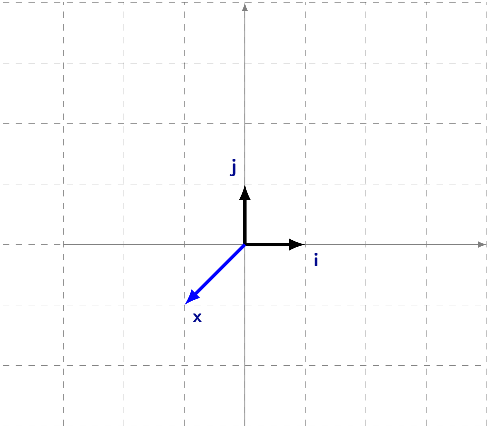
$$\mathbf{x} = \alpha_1 \mathbf{x}_1 + \alpha_2 \mathbf{x}_2 + \ldots + \alpha_m \mathbf{x}_m$$

The standard basis is

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ \ldots \\ \ldots \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ \ldots \\ 0 \end{pmatrix}, \ldots, \begin{pmatrix} 0 \\ \ldots \\ \ldots \\ 0 \\ 1 \end{pmatrix} \right\}$$

$m = n$?

Standard Basis in $\mathbb{R}^2$



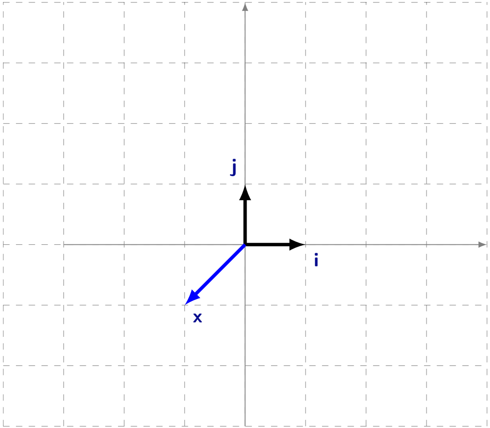
$$\mathbf{i} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\mathbf{j} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$\mathbf{i}$ and $\mathbf{j}$ are **linearly independent** as there is no α such that $\mathbf{i} = \alpha\mathbf{j}$:

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \neq \alpha \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Standard Basis in $\mathbb{R}^2$



Moreover, any $\mathbf{x}$ can be represented as a linear combination of $\mathbf{i}$ and $\mathbf{j}$:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} =$$

$$= x_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + x_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} =$$

$$= x_1 \mathbf{i} + x_2 \mathbf{j}$$

Thus, $(\mathbf{i}, \mathbf{j})$ form a basis in $\mathbb{R}^2$.

Matrix, Definition

An $n \times m$ **matrix** is an array of real numbers distributed in n rows and m columns:

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{pmatrix}$$

Matrices, Typical Structures

A matrix $1 \times m$ is usually called a **row matrix**

A matrix $n \times 1$ is usually called a **column matrix**

A matrix A such that $n = m$ is called a **square matrix**

Null Matrix

The $n \times m$ **null matrix** is the matrix composed entirely by zeros:

$$0_{n\times m} = \begin{pmatrix} 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 0 \end{pmatrix}$$

Identity Matrix

The n-square **identity matrix** is the matrix:

$$I_{n \times n} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

That is, $a_{ij} = 1$ if $i = j$ and $a_{ij} = 0$ otherwise.

Diagonal Matrix

A **diagonal matrix** is an n-square matrix of the form:

$$D = \begin{pmatrix} a_{11} & 0 & \dots & 0 \\ 0 & a_{22} & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & a_{nn} \end{pmatrix}$$

That is, $a_{ij} = 0$ if $i \neq j$.

- Square null matrices and identity matrices are diagonal

Triangular Matrix

An **upper triangular matrix** is an n-square matrix of the form:

$$T = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ 0 & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & a_{nn} \end{pmatrix}$$

That is, $a_{ij} = 0$ if $i > j$.

- Any diagonal matrix is upper triangular

Row Echelon Matrix

An $n \times m$ **row echelon** matrix is of the form:

$$R = \begin{pmatrix} \dots & 0 & a_{1k_1} & \dots & \dots & \dots & \dots & a_{1m} \\ \dots & \dots & \dots & 0 & a_{2k_2} & \dots & \dots & a_{2m} \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & 0 & a_{sk_s} & \dots & a_{sm} \\ 0 & \dots & \dots & \dots & \dots & \dots & \dots & \dots 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & \dots & \dots & \dots & \dots & \dots & \dots & \dots 0 \end{pmatrix}$$

where $k_s > \cdots > k_2 > k_1 \geq 1$

- Upper triangular matrices are row echelon

Symmetric Matrix

An n-square **symmetric** matrix is of the form:

$$S = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{12} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{1n} & a_{2n} & \dots & a_{nn} \end{pmatrix}$$

That is, $a_{ij} = a_{ji}$.

- Square null matrices and identity matrices are symmetric

Matrices: Examples

Example

- $A_1 = \begin{pmatrix} 0 & 0 \end{pmatrix}$ and $A_2 = \begin{pmatrix} 1 & 0 \end{pmatrix}$
- $A_3 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $A_4 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ and $A_5 = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$
- $A_6 = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 1 & 0 \end{pmatrix}$

Addition

The **sum** of two $n \times m$ matrices A and B is defined as:

$$A + B = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1m} \\ b_{21} & b_{22} & \dots & b_{2m} \\ \dots & \dots & \dots & \dots \\ b_{n1} & b_{n2} & \dots & b_{nm} \end{pmatrix} =$$

$$= \begin{pmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \dots & a_{1m} + b_{1m} \\ a_{21} + b_{21} & a_{22} + b_{22} & \dots & a_{2m} + b_{2m} \\ \dots & \dots & \dots & \dots \\ a_{n1} + b_{n1} & a_{n2} + b_{n2} & \dots & a_{nm} + b_{nm} \end{pmatrix}$$

Addition, Examples

- $A_1 + A_2 = \begin{pmatrix} 0 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \end{pmatrix} = A_2$
- $A_3 + A_4 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix}$

Addition, Properties

In general, given $n \times m$ matrices A, B, C:

- $A + 0_{n \times m} = A$
- $A + B = B + A$
- $A + (B + C) = (A + B) + C$

Scalar Multiplication

The **scalar multiplication** of the scalar α and the $n \times m$ matrix A is:

$$\alpha A = \begin{pmatrix} \alpha a_{11} & \alpha a_{12} & \dots & \alpha a_{1m} \\ \alpha a_{21} & \alpha a_{22} & \dots & \alpha a_{2m} \\ \dots & \dots & \dots & \dots \\ \alpha a_{n1} & \alpha a_{n2} & \dots & \alpha a_{nm} \end{pmatrix}$$

We can then define the **difference** of two $n \times m$ matrices:

$$A - B = A + (-1)B = \begin{pmatrix} a_{11} - b_{11} & a_{12} - b_{12} & \dots & a_{1m} - b_{1m} \\ a_{21} - b_{21} & a_{22} - b_{22} & \dots & a_{2m} - b_{2m} \\ \dots & \dots & \dots & \dots \\ a_{n1} - b_{n1} & a_{n2} - b_{n2} & \dots & a_{nm} - b_{nm} \end{pmatrix}$$

Scalar Multiplication, Example

Example

$$3A_3 - A_4 = 3\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + (-1)\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} =$$

$$= \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} + \begin{pmatrix} -1 & 0 \\ -1 & -1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ -1 & 2 \end{pmatrix}$$

Transpose

The **transpose** of an $n \times m$ matrix A is the $m \times n$ matrix A^T that is obtained by interchanging rows and columns, that is:

$$A^T = \begin{pmatrix} a_{11} & a_{21} & \dots & a_{n1} \\ a_{12} & a_{22} & \dots & a_{n2} \\ \dots & \dots & \dots & \dots \\ a_{1m} & a_{2m} & \dots & a_{nm} \end{pmatrix}$$

That is, element a_{ij} becomes element a_{ji}.

Transpose, Example

Example

- $A_1^T = \begin{pmatrix} 0 & 0 \end{pmatrix}^T = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$
- $A_3^T = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}^T = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = A_3$
- $A_6^T = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 1 & 0 \end{pmatrix}^T = \begin{pmatrix} 0 & 0 \\ 0 & 1 \\ 2 & 0 \end{pmatrix}$

Transpose, Properties

- The transpose of a symmetric matrix is itself
- $(A + B)^T = A^T + B^T$
- $(A \cdot B)^T = B^T \cdot A^T$

Inner Product

The **inner product** of a $1 \times m$ row matrix A and a $m \times 1$ column matrix B is:

$$A \cdot B = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \end{pmatrix} \cdot \begin{pmatrix} b_{11} \\ b_{21} \\ \dots \\ b_{m1} \end{pmatrix} = a_{11}b_{11} + a_{12}b_{21} + \cdots + a_{1m}b_{m1}$$

In $\mathbb{R}^n$ we can use terms **inner, scalar, dot** product interchangeably.

Product of Matrices, Preliminaries

The product of two matrices is defined through the inner product. Importantly, and contrary to the case of real numbers:

- The product can only be defined for some pairs of matrices. In particular, $A \cdot B$ is defined only for the case in which A has as many columns as the number of rows of B

Product of Matrices, Definition

The **product** of an $n \times m$ matrix A and an $m \times l$ matrix B is the $n \times l$ matrix:

$$A \cdot B = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{pmatrix} \cdot \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1l} \\ b_{21} & b_{22} & \dots & b_{2l} \\ \dots & \dots & \dots & \dots \\ b_{m1} & b_{m2} & \dots & b_{ml} \end{pmatrix} =$$

$$= \begin{pmatrix} a_{11}b_{11} + \dots + a_{1m}b_{m1} & \dots & a_{11}b_{1l} + \dots + a_{1m}b_{ml} \\ \dots & \dots & \dots \\ a_{n1}b_{11} + \dots + a_{nm}b_{m1} & \dots & a_{n1}b_{1l} + \dots + a_{nm}b_{ml} \end{pmatrix}$$

where the entry ij is the inner product of row i of matrix A and column j of matrix B

Product of Matrices, Examples

- $A_1 \cdot A_6 = \begin{pmatrix} 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & 0 & 2 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \end{pmatrix}$
- $A_3 \cdot A_6 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 & 0 & 2 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 1 & 0 \end{pmatrix} = A_6$
- $A_4 \cdot A_6 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 & 0 & 2 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 1 & 2 \end{pmatrix}$
- What are $A_6 \cdot A_1$, $A_6 \cdot A_3$ and $A_6 \cdot A_4$?

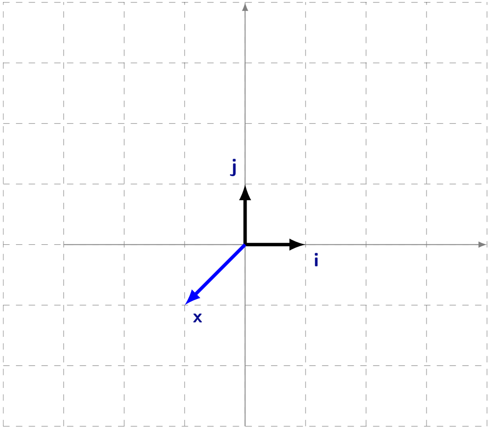
Product of Matrices, Properties

- Given an $n \times m$ matrix the product by the identity matrix (either premultiplying by the $n \times n$ identity or post-multiplying by the $m \times m$ identity) leaves A unaltered, that is $I_{n\times n} \cdot A = A \cdot I_{m\times m} = A$.
- When they are well-defined: $(A \cdot B) \cdot C = A \cdot (B \cdot C)$
- When they are well-defined: $A \cdot (B + C) = A \cdot B + A \cdot C$
- If they exist, the product $A \cdot B$ may be different to the product $B \cdot A$

Product, Intuition

A **linear transformation** between $\mathbb{R}^n$ and $\mathbb{R}^m$ is a mapping $\mathbf{x} \to A\mathbf{x}$.

$\mathbb{R}^2$



$$\mathbf{i} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

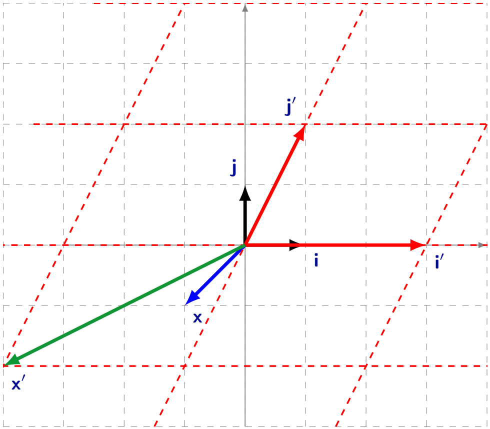
$$\mathbf{j} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\mathbf{x} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$$

Thus,

$$\mathbf{x} = -\mathbf{i} - \mathbf{j}$$

Product, Intuition



$$A_5 = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$$

$$\mathbf{i}' = A_5\mathbf{i} = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$$

$$\mathbf{j}' = A_5\mathbf{j} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

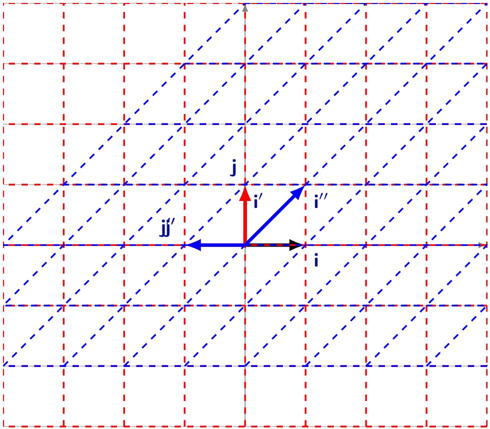
$$\mathbf{x}' = A_5\mathbf{x} = \begin{pmatrix} -4 \\ -2 \end{pmatrix}$$

Note that

$$\mathbf{x}' = -\mathbf{i}' - \mathbf{j}'$$

Actually any $\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$ is transformed to $\mathbf{y}' = y_1\mathbf{i}' + y_2\mathbf{j}'$

Product, Order Matters



Basis:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Rotation:

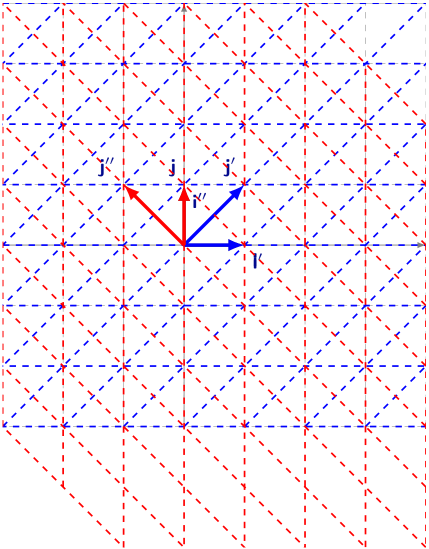
$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

Shear:

$$S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

Result: $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \left(\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}$

Product, Order Matters



Basis:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Shear:

$$S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

Rotation

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

Result: $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \left(\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$

Inverse, Definition

The inverse operation can only be defined for square matrices!

Definition

The **inverse** *of an* n*-square matrix* A *is the* n*-square matrix, denoted* A^{-1}*, such that*
$A^{-1} \cdot A = A \cdot A^{-1} = I$

Inverse, Example

Example

The inverse of A_4 is $A_4^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$ because

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

Inverse, Properties

- The inverse does not necessarily exist
- Whenever it exists, it is unique
- $(A \cdot B)^{-1} = B^{-1} \cdot A^{-1}$

Linear Systems, Definition

A **system of n linear equations** is of the form

$$\begin{gathered} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1m}x_m = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2m}x_m = b_2 \\ \cdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nm}x_m = b_n \end{gathered}$$

It can be written through an $n \times m$ matrix A, an $m \times 1$ column matrix x and an $n \times 1$ column matrix b, as:

$$A \cdot \mathbf{x} = \mathbf{b}$$

Linear Systems, Remarks

Linear systems must be of one of these types:

- Having exactly one solution
- Having no solution
- Having infinitely many solutions

Whenever $b_1 = b_2 = \cdots = b_n = 0$, the system is called **homogeneous**. Homogeneous systems must have either one or infinite solutions. The reason is that $x_1 = x_2 = \cdots = x_m = 0$ is always a solution

Linear Systems, Solution

Suppose that A is a square matrix. Notice that whenever A is invertible, we can simply pre-multiply the system by A^{-1}, to obtain:

$$A^{-1} \cdot A \cdot \mathbf{x} = I\mathbf{x} = A^{-1}\mathbf{b}$$

which provides immediately the (unique) values of x that solve the system.

(Cramer's rule is sometimes used to facilitate solutions)

Linear Systems, Examples

Example 1 Consider the system $A_4 \cdot \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = A_2^T$, i.e.,

$$x_1 = 1$$

$$x_1 + x_2 = 0$$

Since $A_4^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$, we have

$$A_4^{-1} A_4 \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = I \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} =$$

$$= A_4^{-1} A_2^T = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Linear Systems, Examples

Example 2 Consider the system $A_5 \cdot \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} -4 \\ -2 \end{pmatrix}$, Ie.,

$$3x_1 + x_2 = -4$$

$$2x_2 = -2$$

Since $A_5^{-1} = \begin{pmatrix} 1/3 & -1/6 \\ 0 & 1/2 \end{pmatrix}$, we have

$$A_5^{-1} A_5 \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = I \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = A_5^{-1} \begin{pmatrix} -4 \\ -2 \end{pmatrix} =$$

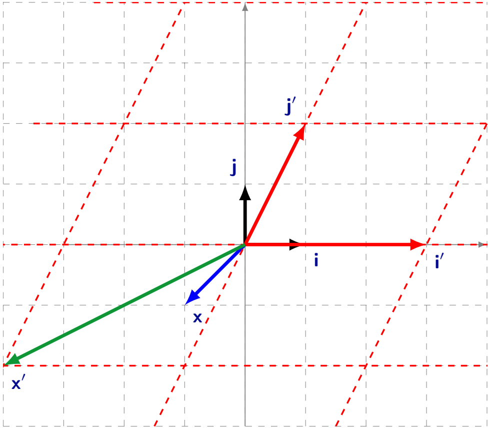
$$= \begin{pmatrix} 1/3 & -1/6 \\ 0 & 1/2 \end{pmatrix} \cdot \begin{pmatrix} -4 \\ -2 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$$

Linear Systems, Invertibility

- Invertibility is equivalent to having one solution
- Non-invertibility is equivalent to having none or infinite solutions

Appendix: Gauss-Jordan Elimination

Linear Systems, Illustration



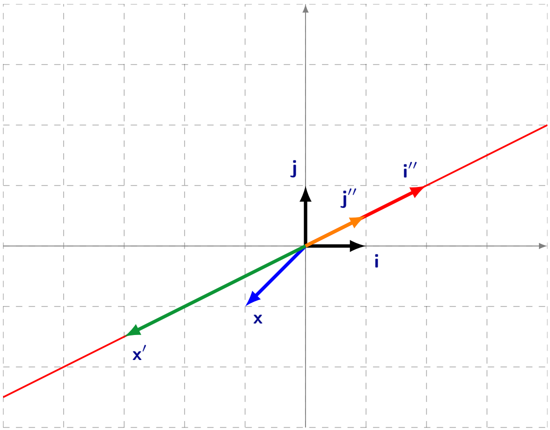
Recall **Example 2**:

$$A_5 \cdot \mathbf{x} = \mathbf{x}'$$

Thus, we need to apply the inverse transformation, hence inverting A_5:

$$\mathbf{x} = A_5^{-1} \cdot \mathbf{x}'$$

Linear Systems, Illustration



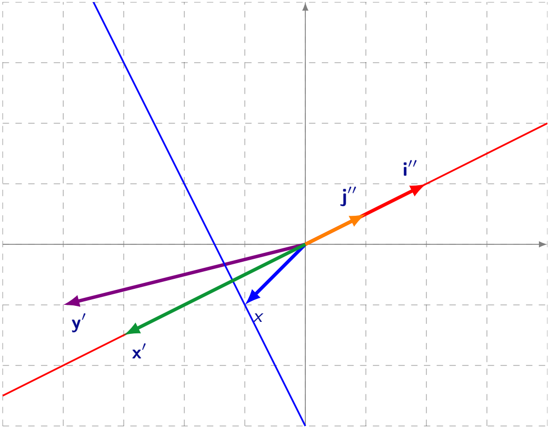
$$B = \begin{pmatrix} 2 & 1 \\ 1 & 1/2 \end{pmatrix}$$

$$\mathbf{i}' = B \cdot \mathbf{i} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$\mathbf{j}' = B \cdot \mathbf{j} = \begin{pmatrix} 1 \\ 1/2 \end{pmatrix}$$

$$\mathbf{x}' = B \cdot \mathbf{x} = \begin{pmatrix} -3 \\ -3/2 \end{pmatrix}$$

Linear Systems, Illustration



$$B = \begin{pmatrix} 2 & 1 \\ 1 & 1/2 \end{pmatrix}$$

All vectors are mapped on the same line. Thus, any $\mathbf{y} = \begin{pmatrix} \alpha \\ -3 - 2\alpha \end{pmatrix}$ solves $B \cdot \mathbf{y} = \mathbf{x}'$ and there is no $\mathbf{y}$ which solves

$$B \cdot \mathbf{y} = \mathbf{y}' = \begin{pmatrix} -4 \\ -1 \end{pmatrix}$$

Indeed, $B \cdot \mathbf{y} = \begin{pmatrix} 2y_1 + y_2 \\ y_1 + 1/2y_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} y$, so all must belong to the same line

Determinant, Definition

The determinant of square matrices is defined recursively:

- If $A = (a)$, then

$$\det A = a$$

- If $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$, then

$$\det A = a_{11}a_{22} - a_{12}a_{21}$$

Determinant, Definition

$$A = \begin{pmatrix} \boxed{a_{11}} & \boxed{a_{12}} & \dots & \boxed{a_{1n}} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix}$$

$$\det A = a_{11}M_{11} - a_{12}M_{12} + \ldots + (-1)^{n+1}a_{1n}M_{1n}$$

where M_{ij} is a **minor** of the matrix ($C_{ij} = (-1)^{i+j}M_{ij}$ is called the cofactor), which is simply the determinant of the reduced matrix obtained by eliminating row i and column j of the matrix.

Determinant, Remarks

In general:

$$\det A = \sum_{i=1}^{n} a_{ji} C_{ji}$$

so we can use any row j (or column) to define it

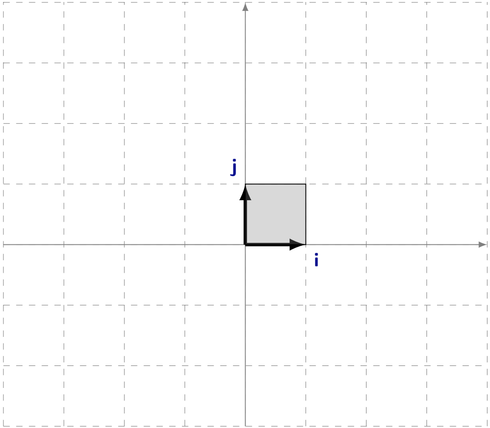
- $\det A^T = \det A$
- $\det A \cdot B = \det A \det B$
- The determinant of a triangular matrix is simply the product of its diagonal entries

Determinant, Application

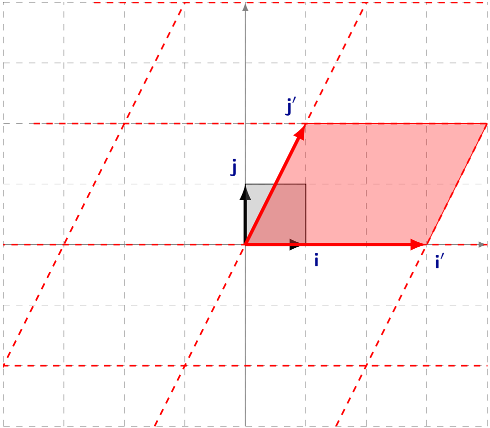
The n-square matrix A is invertible if and only if $\det A \neq 0$

The linear system $A \cdot \mathbf{x} = \mathbf{b}$, with n equations and n unknowns, has a unique solution if and only if $\det A \neq 0$

Determinant, Illustration



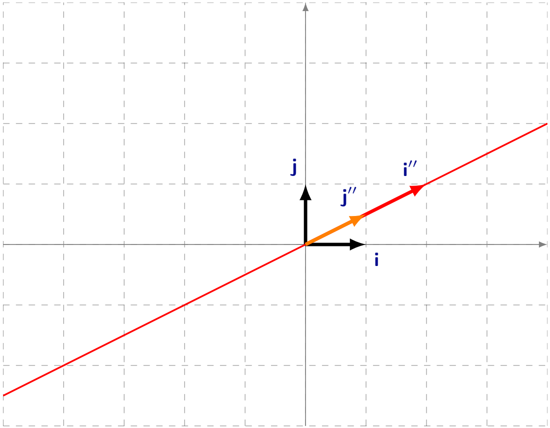
Positive Determinant, Illustration



$$A = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$$

$$\det A = 6$$

Zero Determinant, Illustration

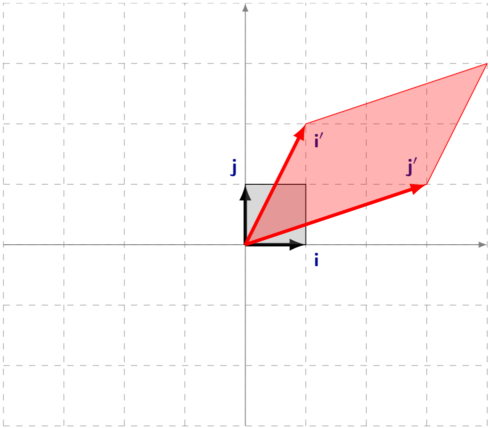


$$B = \begin{pmatrix} 2 & 1 \\ 1 & 1/2 \end{pmatrix}$$

$$\det B = 0$$

Matrix B does not have the inverse, so it collapses $\mathbb{R}^2$ into a line, so the determinant equals to zero. Clearly the system of equations $B \cdot \mathbf{y} = \mathbf{y}'$ is not generically solvable, and if it is, then it has infinitely many solutions.

Negative Determinant, Illustration



$$C = \begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix}$$

$$\det C = -5$$

Note the flip of the axes.

Rank

The **rank** of A is the maximal number of linearly independent rows or columns.

We say that an $n \times m$ matrix is of **full rank** if $rank\, A = \min\{n, m\}$.

Some properties

- $rank\, A = rank\, A^T$
- $rank\, A \leq$ number of rows
- $rank\, A \leq$ number of columns

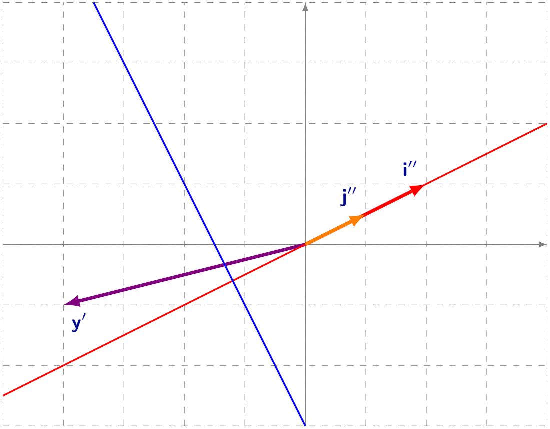
Rank, Linear Systems

- A system of linear equations $A \cdot \mathbf{x} = \mathbf{b}$ has solutions if and only if the rank of the matrix of coefficients A is equal to the rank of the augmented matrix $(A, \mathbf{b})$

$$(A, \mathbf{b}) = \begin{pmatrix} a_{11} & \dots & a_{1m} & b_1 \\ \vdots & \ddots & \vdots & \vdots \\ a_{n1} & \dots & a_{nm} & b_m \end{pmatrix}$$

- The linear system $A \cdot \mathbf{x} = \mathbf{b}$, with n equations and n unknowns, has a unique solution if and only if $det(A) \neq 0$, i.e. if and only if it has full rank
- Thus, if system $A \cdot \mathbf{x} = \mathbf{b}_1$ has a unique solution, then any system $A \cdot \mathbf{x} = \mathbf{b}_2$ has a unique solution.

Rank, Illustration



$$B = \begin{pmatrix} 2 & 1 \\ 1 & 1/2 \end{pmatrix}$$

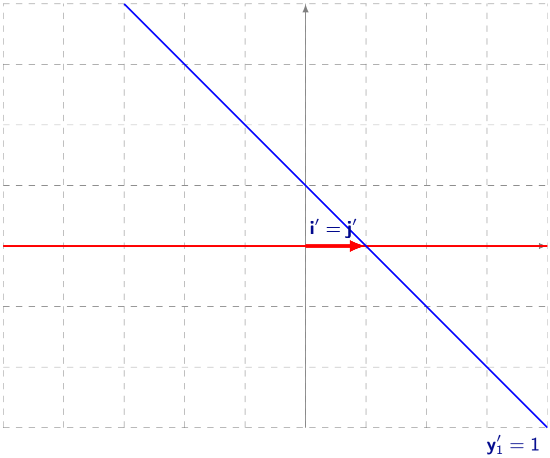
$$2y_1 + y_2 = y_1'$$

$$y_1 + 1/2 y_2 = y_2'$$

rank $(B, \mathbf{y}') = 2 >$ *rank* B if $\mathbf{y}' \neq (2a, a)^T$, so system $B \cdot \mathbf{y} = \mathbf{y}'$ does not have a solution

rank $(B, \mathbf{y}') = 1 =$ *rank* B if $\mathbf{y}' = (2a, a)^T$, so system $B \cdot \mathbf{y} = \mathbf{y}'$ has infinitely many solutions in the form $\mathbf{y} = (\alpha, -3 - 2\alpha)^T$

Rank, Illustration



$$D = \begin{pmatrix} 1 & 1 \end{pmatrix}$$

$$y_1 + y_2 = y_1'$$

rank $D =$ *rank* $(D, \mathbf{y}') = 1$, full rank, so system $D \cdot \mathbf{y} = \mathbf{y}'$ always has a solution, in fact infinitely many $\mathbf{y} = (a, y_1' - a)$.

Trace

The **trace** of a square matrix A is the sum of the elements of its diagonal.

$$tr(A) = \sum_i a_{ii}$$

Example

$$tr \begin{pmatrix} 0 & 2 & 4 \\ 1 & -2 & -3 \\ -1 & 1 & 2 \end{pmatrix} = 0$$

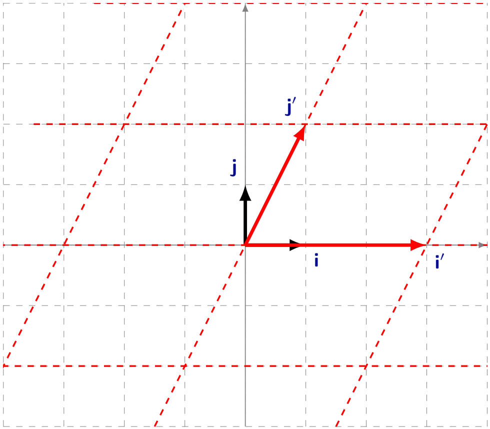
Span

The **span** of a collection of n-dimensional vectors, $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_m$, is the set of all their linear combinations:

$$span[\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_m] = \{\mathbf{x} : \mathbf{x} = \alpha_1 \mathbf{x}_1 + \alpha_2 \mathbf{x}_2 + \cdots + \alpha_m \mathbf{x}_m\}$$

- we need at least n vectors to span $\mathbb{R}^n$
- a collection of n-dimensional vectors, $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_m$ spans $\mathbb{R}^n$ if and only if $rank\ X = (\mathbf{x}_1, \mathbf{x}_2, \ldots \mathbf{x}_m) = n$
- **basis** is a minimal collection of vectors which spans $\mathbb{R}^n$

Span, Illustration



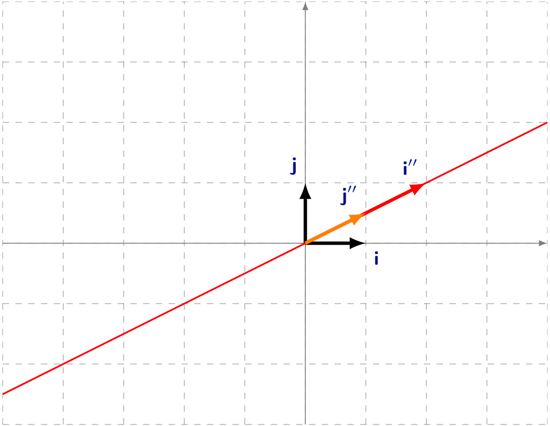
Both $(\mathbf{i}, \mathbf{j})$ and $(\mathbf{i}', \mathbf{j}')$ span $\mathbb{R}^2$:

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = x_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + x_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

and

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \frac{1}{3}\left(x_1 - \frac{x_2}{2}\right) \begin{pmatrix} 3 \\ 0 \end{pmatrix} + \frac{x_2}{2} \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

Span, Illustration



Vectors

$$\begin{pmatrix} 2 \\ 1 \end{pmatrix} \text{ and } \begin{pmatrix} 1 \\ 1/2 \end{pmatrix}$$

do not span $\mathbb{R}^2$, they only span a one-dimensional space.

Span and Linear Systems

Not surprisingly, there is an intimate connection between linear combinations/independence and systems of equations.

- Vector $\mathbf{b}$ is a linear combination of vectors $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_m$ (Ie., it belongs to their span or it is spanned by these vectors) if and only if the linear system $X\mathbf{y} = \mathbf{b}$ has at least one solution
- Linear independence is equivalent to $\alpha_1 \mathbf{x}_1 + \ldots \alpha_m \mathbf{x}_m = 0$ implies $\alpha_1 = \cdots = \alpha_m = 0$, and this can be written as the homogeneous system $X \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \ldots \\ \alpha_m \end{pmatrix} = 0$ has at most one solution

Span, Further Results

Some results:

- (All) bases of $\mathbb{R}^n$ have n vectors
- A set of n vectors is a basis if they are linearly independent
- A linearly independent collection of vectors is a basis if adding any other vector makes the collection dependent
- A set that spans $\mathbb{R}^n$ is a basis if removing any vector makes the remaining collection not to span $\mathbb{R}^n$
- Every spanning set contains a basis
- Every set of independent vectors can be extended to a basis

Basis Transformation

Let B and D be n-square matrices that correspond to two basis of $\mathbb{R}^n$. Then, we can find an invertible matrix K such that $B \cdot K = D$.

Just let $K = B^{-1} \cdot D$. Notice that

- $B \cdot B^{-1} \cdot D = I \cdot D = D$
- $det(B^{-1} \cdot D) = det(B^{-1})det(D) \neq 0$.

There is a clear connection between linear independence or span and the determinant or the rank!

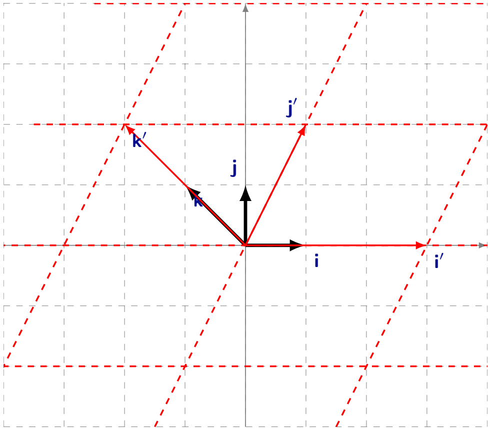
Eigenvalues and Eigenvectors

An **eigenvalue** and an **eigenvector** of an n-square matrix A are a scalar λ and an n-dimensional vector $\mathbf{x}$ such that $A \cdot \mathbf{x} = \lambda \mathbf{x}$

The eigenvalues of A are the roots of the **characteristic polynomial**, i.e., the determinant of matrix $A - \lambda I$. There are n roots, and some may be complex or repeated.

Note, that if $A \cdot \mathbf{x} = \lambda \mathbf{x}$ then $A \cdot (\alpha \mathbf{x}) = \lambda(\alpha \mathbf{x})$, so eigenvectors are defined up to a multiple.

Eigenvectors, Illustration



$$A = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$$

so

$$\det(A - \lambda I) = \begin{vmatrix} 3-\lambda & 1 \\ 0 & 2-\lambda \end{vmatrix} = 0$$

and thus $\lambda_1 = 3, \lambda_2 = 2$.

$$\begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 3 \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

so $(1,0)^T$ is an eigenvector - indeed $\mathbf{i}$ stays on the same line!

$$\begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 2 \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Gives an eigenvector of $\mathbf{k} = (-1,1)^T$.

Eigenvectors, Example

Example

$det \begin{pmatrix} 1-\lambda & 0 \\ 1 & 1-\lambda \end{pmatrix} = (1-\lambda)^2 = 0$ leads to $\lambda = 1$, repeated

The eigenvectors are all vectors such that

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

That is, vectors with $x_1 + x_2 = x_2$, or simply, $x_1 = 0$, e.g. $(0, 1)^T$.

Eigenvalues, Properties

- All the eigenvalues of a symmetric matrix are real
- Eigenvalues of non-symmetric matrices are not necessarily real (but complex)
- $\det A = \lambda_1 \lambda_2 \dots \lambda_n$ (recall our linear transformation)
- $tr(A) = \lambda_1 + \lambda_2 + \cdots + \lambda_n$

Eigenvalues, Some Remarks

Although eigenvalues and eigenvectors are usually tedious, 2-square matrices are simple.

- If $\det A < 0$, eigenvalues have opposite sign
- If $\det A > 0$, eigenvalues have same sign
  - If $tr(A) > 0$, they are positive
  - If $tr(A) < 0$, they are negative

Eigenvalues, Some Remarks

If a matrix has all the eigenvalues real and different, the eigenvectors are independent and span $\mathbb{R}^n$. Also, they form the columns of an invertible matrix V such that, denoting D the diagonal matrix formed by the eigenvalues, $V^{-1} \cdot A \cdot V = D$

Notice that $A = V \cdot D \cdot V^{-1}$ and $A^n = V \cdot D^n \cdot V^{-1}$

Can you use this property for inverting a matrix?

Negative/Positive Definite Martices

An $n \times n$ matrix A is called positive (negative) definite, if for all $\mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ $\mathbf{x}^T \cdot A \cdot \mathbf{x} > (<) 0$.

An $n \times n$ matrix A is called positive (negative) semi-definite, if for all $\mathbf{x} \in \mathbb{R}^n$ $\mathbf{x}^T \cdot A \cdot \mathbf{x} \geq (\leq) 0$.

Negative semi-definite matrices are of particular interest in multivariate optimization – they arise from second order conditions. One can use the Sylvester criterion to check whether a **symmetric** matrix A is negative semi-definite. It is, if corner minors change sign:

$$\begin{vmatrix} a_{11} \end{vmatrix} < 0, \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} > 0, \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} < 0, \ldots$$

Matrix A is negative semi-definite if all its eigenvalues are non-positive. Analogous result applies for positive (semi-)definite matrices.

Appendix: Gauss-Jordan Elimination

In general, some pre-multiplication of A and b by specific matrices may allow to write:

$$E \cdot A \cdot \mathbf{x} = R \cdot \mathbf{x} = E \cdot \mathbf{b}$$

where R is row echelon, and provide a simple method to solve the system. This kind of transformation into a row echelon matrix is always possible, and it is known as Gauss elimination or Gauss-Jordan elimination (depending on the operations performed).

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Appendix: Gauss-Jordan Elimination

Gauss elimination operations:

- Swapping two rows (pre-multiplying by identity, with same operation)
- Example: $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$

Appendix: Gauss-Jordan Elimination

Gauss elimination operation:

- Adding k times one row to another row (pre-multiplying by identity, with same operation)
- Example: $\begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

Appendix: Gauss-Jordan Elimination

Gauss-Jordan elimination operation:

- Multiplying a row by a non-zero scalar (pre-multiplying by identity, with same operation)

Appendix: Gauss-Jordan Elimination

Example

$$2x_2 + 4x_3 = -8$$

$$x_1 - 2x_2 - 3x_3 = 0$$

$$-x_1 + x_2 + 2x_3 = 3$$

Appendix: Gauss-Jordan Elimination

Example

$$\left(\begin{array}{ccc|c} 0 & 2 & 4 & -8 \\ 1 & -2 & -3 & 0 \\ -1 & 1 & 2 & 3 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & -2 & -3 & 0 \\ 0 & 2 & 4 & -8 \\ -1 & 1 & 2 & 3 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} 1 & -2 & -3 & 0 \\ 0 & 2 & 4 & -8 \\ 0 & -1 & -1 & 3 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & -2 & -3 & 0 \\ 0 & 2 & 4 & -8 \\ 0 & 0 & 1 & -1 \end{array}\right)$$

$x_3 = -1$, $x_2 = -2$ and $x_1 = -7$

Appendix: Gauss-Jordan Elimination

Example

$$2x_1 + x_2 + 2x_3 = -12$$

$$4x_1 + 3x_2 - 3x_3 = -2$$

$$6x_1 + 5x_2 - 8x_3 = 9$$

Appendix: Gauss-Jordan Elimination

Example

$$\left(\begin{array}{ccc|c} 2 & 1 & 2 & -12 \\ 4 & 3 & -3 & -2 \\ 6 & 5 & -8 & 9 \end{array}\right) \sim \left(\begin{array}{ccc|c} 2 & 1 & 2 & -12 \\ 0 & 1 & -7 & 22 \\ 6 & 5 & -8 & 9 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} 2 & 1 & 2 & -12 \\ 0 & 1 & -7 & 22 \\ 0 & 2 & -14 & 45 \end{array}\right) \sim \left(\begin{array}{ccc|c} 2 & 1 & 2 & -12 \\ 0 & 1 & -7 & 22 \\ 0 & 0 & 0 & 1 \end{array}\right)$$

No solution

Appendix: Gauss-Jordan Elimination

Example

$$x_1 + 2x_2 + 2x_3 = 0$$

$$2x_1 + 4x_2 + 6x_3 = 0$$

$$3x_1 + 6x_2 + 8x_3 = 0$$

Appendix: Gauss-Jordan Elimination

Example

$$\left(\begin{array}{ccc|c} 1 & 2 & 2 & 0 \\ 2 & 4 & 6 & 0 \\ 3 & 6 & 8 & 0 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 2 & 2 & 0 \\ 0 & 0 & 2 & 0 \\ 3 & 6 & 8 & 0 \end{array}\right) \sim$$

$$\sim \left(\begin{array}{ccc|c} 1 & 2 & 2 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 2 & 0 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 2 & 2 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Infinite solutions: $x_3 = 0$, $x_1 = -2x_2$, with $x_2 \in \mathbb{R}$

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Appendix: Gauss-Jordan Elimination and Rank

Example

- $\begin{pmatrix} 0 & 2 & 4 \\ 1 & -2 & -3 \\ -1 & 1 & 2 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & -3 \\ 0 & 2 & 2 \\ 0 & 0 & 1 \end{pmatrix}$, full rank
- $\begin{pmatrix} 2 & 1 & 2 \\ 4 & 3 & -3 \\ 6 & 5 & -8 \end{pmatrix} \sim \begin{pmatrix} 2 & 1 & 2 \\ 0 & 1 & -7 \\ 0 & 0 & 0 \end{pmatrix}$, not full rank
- $\begin{pmatrix} 1 & 2 & 2 \\ 2 & 4 & 6 \\ 3 & 6 & 8 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 2 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$, not full rank

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